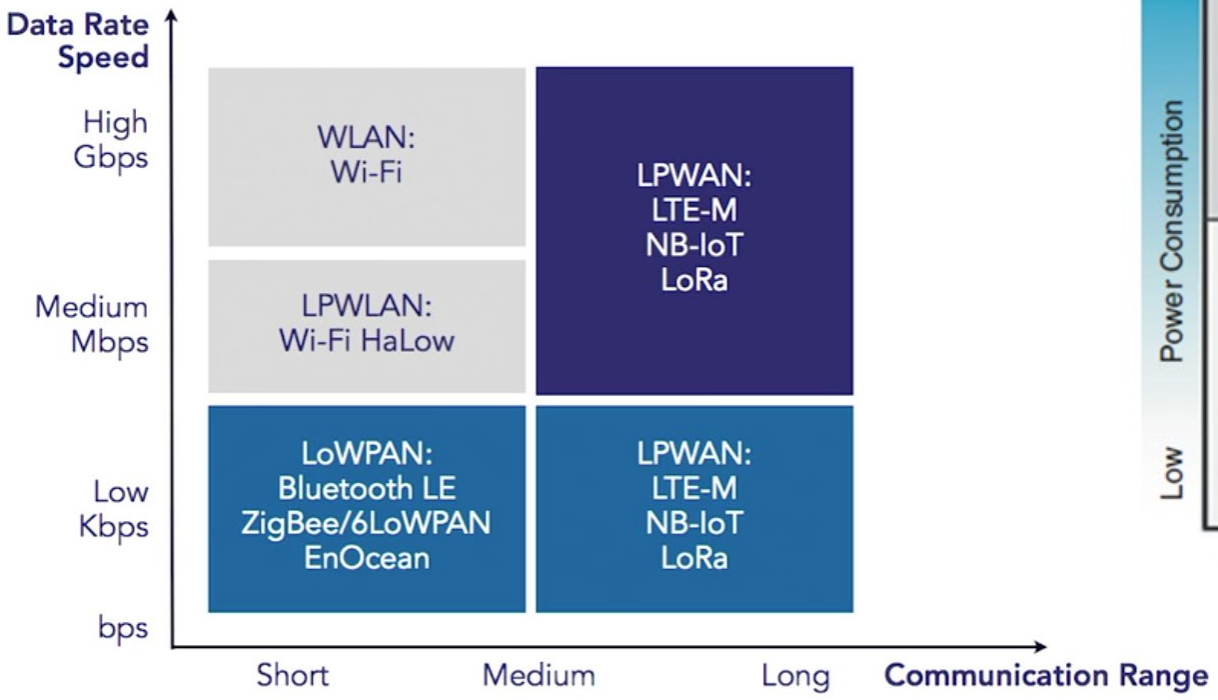
Design of Enterprise IoT

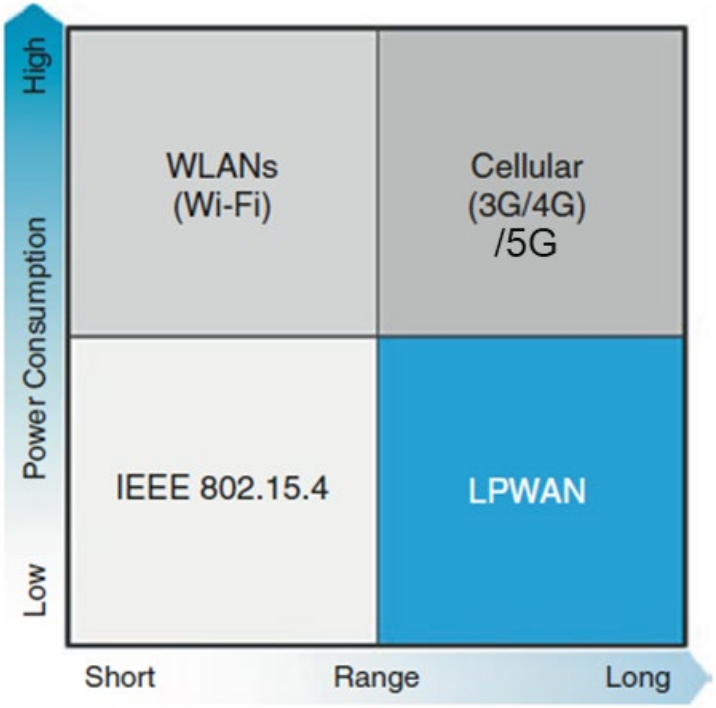
Tutorial 3

IoT Network Connectivity

Determining IoT Network Technologies

The choice of IoT network technologies should be determined from several key perspectives, including **device power consumption**, **communication range**, **data rate**, **latency**, **reliability**, scalability, security, deployment environment, network topology, and cost.





Technical Comparison: IoT Network Technologies

Technology	Comm. Range	Data Rate	Latency	Power Consumption	Network Category
Ethernet	100 m (wired) per segment	> 1 Gbps	< 1 ms	High to medium	Wired Network
Wi-Fi	50 ~ 100 m	> 100 Mbps	< 10 ms	High	WLAN
4G (LTE)	5 ~ 20 km (area served by a single cellular-based station)	> 100 Mbps	30 ~ 50 ms	High	Cellular Network
IEEE 802.11ah (Wi-Fi HaLow)	1 km	150 kbps ~ 1 Mbps	30 ~ 100 ms	Low	LPWLAN
LTE-M	10 ~ 15 km	< 1 Mbps	10 ~ 100 ms	Low	LPWAN
IEEE 802.15.4	10 ~ 100 m	< 250 kbps	20 ms	Very Low	LoWPAN
ZigBee / 6LoWPAN	10 ~ 100 m	< 250 kbps	20 ~ 50 ms	Very Low	LoWPAN
NB-IoT	15 ~ 20 km	20 ~ 250 kbps	1.6 ~ 10s	Very Low	LPWAN
LoRa	15 ~ 20 km	0.3 ~ 50 kbps	Seconds	Ultra Low	LPWAN

An Example of IoT Connectivity

Smart Irrigation - An Illustrative Overview

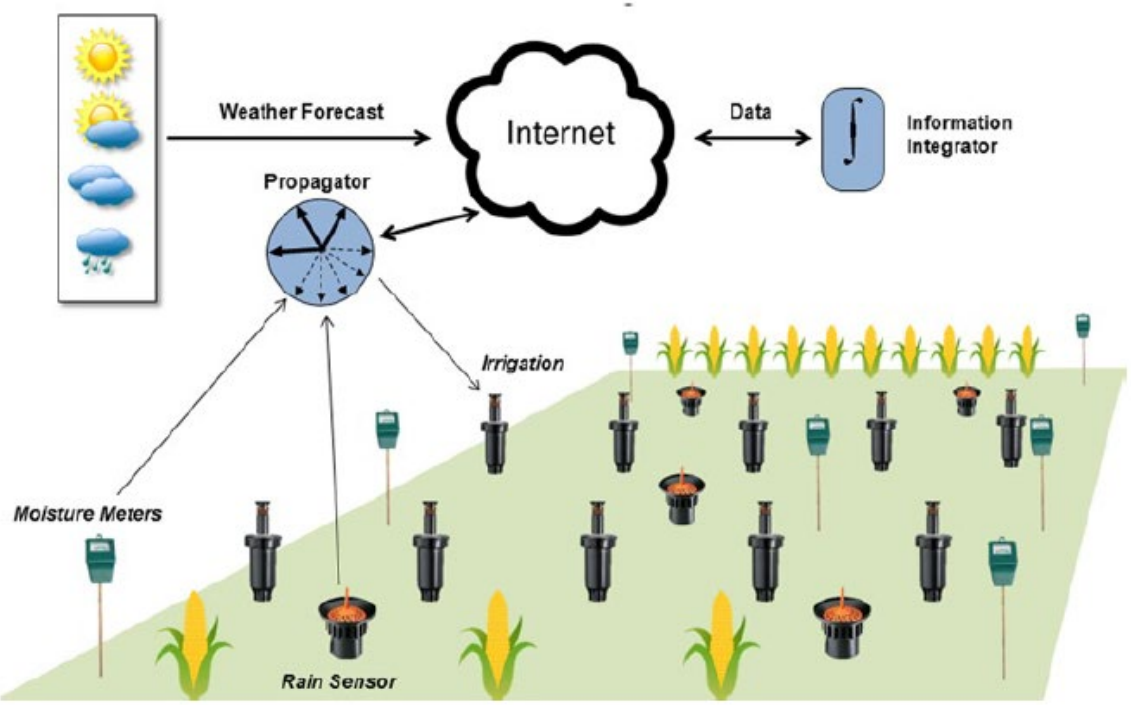


Figure 2-10. *An integrator function retrieves data from end devices such as moisture sensors and external feeds such as the expected precipitation and humidity, using the information to control irrigation valves*

Source: (Decosta, 2013)

- What devices are used?
  - **Moisture sensor**: measure soil moisture level, location of sensing area
    - Example product: Wireless Soil Moisture Sensor (https://www.pasco.com/products/sensors/wireless/ps-3228)
  - **Rain sensor**: detect the rain
    - Whenever it rains, the sensors are activated, which turns off the irrigation controller almost immediately.
    - Example product: Rainfall gauge sensor (https://www.maherelectronica.com/en/agricultural-climate-sensors/rainfall-gauge-sensor-for-agriculture/)
  - **Irrigation sprinkler**: actuate to irrigate
    - Example product: BERMAD Electronic Auto Metering Valve (https://www.bermad.com/product/ir-910-me-kx/)
  - *Weather station for agriculture: provide weather forecast*
    - *Example product: Environdata (https://www.environdata.com.au/industry-specific-weather-stations/agriculture-weather-stations)*
  - *Agriculture drone*
    - *Scan the crop field and collect a more global view of the terrain*
    - *Check irrigation status of the soil and monitor water needs (e.g., via infra-red technologies)*
    - *Example product: XAG drone (https://www.xagaustralia.com.au/agriculture-drone-benefits)*

- Do you need to connect these devices to each other? Why and how?
  - Yes, all three devices need to be connected
    - Information captured by both the **moisture sensors** and **rain sensors** needs to be collected for real-time monitoring of soil moisture status
    - The **irrigation sprinklers** need to act on the above information. For example, if the moisture level drops below a preset value, irrigation should be performed immediately; and when rainfall conditions are detected, irrigation should be switched off.
  - All three devices are connected through **Propagators**
    - Each propagator is used to connect a network of moisture sensors, rain sensors and irrigation sprinklers within a designated area in crop field.
      - A propagator (acting as a router) receives data originating from the connected devices and decides the direction of data transmission based on a set of rules.
      - A propagator (acting as a gateway) processes the received data to extract contextual information from sensors, e.g., IPv6 address and location information, which allows a more precise location and identification of specific individual sensors
      - A propagator (acting as a fog node) analyses the received data to decide the need for irrigation and sends control commands to irrigation sprinklers.

- How do you connect these devices to network?
  - What communication/network technologies are suitable to apply?

    [Option1] **Wired network** technology such as **Ethernet** is applied to connect all moisture/rain sensors and irrigation sprinklers. It is more cost-effective, and the use of wired network connections ensures a stable and reliable communication link between devices.

    [Option2] **Wireless network** technology such as **LoRaWAN** is applied. All devices are of low power consumption and require medium-long distance in crop field. Both NB-IoT and LoRaWAN are for LPWAN. For NB-IoT, there is a monthly subscription fee (Feng et al., 2019). LoRaWAN usually has a maintenance cost but is preferred as a wireless network technology in such context.

    In addition, **Zigbee** can be another option, it is among the lowest power consumption and has better performance in real-time response, but its coverage is limited. Hence, the adoption of this option depends on the communication range requirement in crop field.

Tutorial Activity

IoT Use Case: Intelligent Port System for Fraud Detection



Photo credit to Smart Port Solution

Please read the use case and discuss selected network technologies for the proposed solution

Intelligent Port System for Fraud Detection

Port operations play a crucial role in global trade, handling vast amounts of cargo daily. However, fraudulent activities, such as cargo theft, pose significant financial and operational risks. According to China Police Daily, between 2013 and 2017, Chinese Public Security Traffic Management Authorities cracked over 4,000 criminal cases related to port logistics (Song et al., 2020) . Cargo theft is often considered a "silent crime" because losses frequently go unreported and may only be discovered weeks or even months later.

One common cargo theft tactic involves manipulating a truck's empty weight before entering the port. Fraudsters add materials like liquid tanks, cement, or scrap to artificially increase the truck's weight during the initial weigh-in. After getting the empty truck weighed, these materials are secretly removed, creating a weight discrepancy that allows the truck to load excess cargo undetected. Such fraudulent practices lead to financial losses, regulatory violations, and operational inefficiencies.

Traditional fraud detection methods rely heavily on manual inspections, post-event investigations, and rule-based monitoring systems, which struggle to detect anomalies in real time. These limitations highlight the need for an intelligent port system that enhances security, improves efficiency, and minimizes fraudulent activities. IoT technology offers a promising solution by enabling real-time tracking, automated data collection, and intelligent analytics, allowing ports to proactively detect fraud and optimise operations.

Song, R., Cui, W., Vanthienen, J., Huang, L., & Wang, Y. (2022). Business process redesign towards IoT-enabled context-awareness: the case of a Chinese bulk port. *Business Process Management Journal*, *28*(3), 656-683.

Proposed IoT Solution

An IoT-based intelligent port system leverages advanced sensors, connectivity, and data analytics to enhance security and efficiency. Key components include:

- **Smart Weighbridge Systems:** Infrared sensors and weight sensors detect improper truck positioning and compare real-time weight with historical records, ensuring accuracy in cargo measurement.
- **Environmental Context Integration:** Weather sensors provide real-time rainfall data to detect fraudulent weight manipulation involving hidden water tanks.
- **GPS-Based Route Monitoring:** GPS tracking ensures trucks follow designated routes, detecting unauthorised stops or route deviations that may indicate fraud.
- **IoT-Enabled Cargo Surveillance:** Smart cameras, RFID scanners, and environmental sensors verify cargo handling, preventing unauthorised cargo movement.

Activity

Start with the list of sensors and devices you selected last week for the above Intelligent Port System for the Fraud Detection scenario.

- Determine if the selected devices need to connect with each other and to the network. Justify why connectivity is necessary.
- Determine which communication/network technologies (including protocols) can be used to connect the sensors to the network.

  [Hint: Consider the following questions when making your decision]

  - What are the range and coverage requirements?
    - E.g., what is the required communication distance?
  - What are the power consumption constraints?
    - E.g., are the devices battery-powered or connected to a power source
  - What is the required data rate (how much data needs to be transmitted over time)?
  - What are the latency requirements (how fast does data need to be transmitted)?
    - E.g., is real-time or near-real-time communication needed?
  - What are the security and reliability requirements?
  - What are the cost considerations?
    - E.g., what are the budget constraints?

Access control system



Smart Weighbridge System





GPS tracking





